Waste Management Analytics System

Step-by-step setup guide for Windows, macOS, and Linux

1. Overview & Requirements

This project is an interactive municipal solid-waste analytics dashboard powered by Streamlit, SQLite, and Plotly. It includes automated data validation, key indicator calculations, state and year comparisons, priority classification, and PDF/CSV/ZIP report exports.

Before you start, please ensure:

- Python 3.11 or newer is installed on your computer (download from https://www.python.org).
- On Windows: Make sure to check the box 'Add python.exe to PATH' during installation.
- An active internet connection is required only once to install the Python libraries.
- No database server, external API keys, or cloud accounts are needed.

2. Instructions for Windows (Command Prompt / PowerShell)

After extracting the received ZIP file, follow these quick steps:

Step	Action / Exact Command
Step 1: Extract	Right-click the ZIP file and select 'Extract All...'. Choose a simple path like Desktop.
Step 2: Terminal	Open the unzipped folder. Click the address bar, type cmd or powershell, and press Enter.
Step 3: Virtual Env	Create a clean virtual environment: > python -m venv .venv
Step 4: Install	Install the required packages: > .venv\Scripts\python.exe -m pip install -r requirements.txt
Step 5: Run App	Start the dashboard: > .venv\Scripts\python.exe -m streamlit run app/streamlit_app.py
Step 6: Open Browser	Open your web browser and go to: http://localhost:8501

3. Instructions for macOS and Linux

Open your Terminal application, navigate to the extracted folder, and run:

Step	Terminal Command
Step 1: Navigate	cd /path/to/extracted_folder
Step 2: Virtual Env	python3 -m venv .venv
Step 3: Install	.venv/bin/python -m pip install -r requirements.txt
Step 4: Run App	.venv/bin/streamlit run app/streamlit_app.py
Step 5: Open Browser	Open http://localhost:8501 in Chrome, Safari, or Firefox.

4. Key Commands & Troubleshooting

Topic	Solution / Details
How to Stop App	Press Ctrl + C in the terminal window where Streamlit is running.
Port 8501 Busy	Add --server.port 8502 at the end of the run command and visit http://localhost:8502.
Rebuild Database	The SQLite database (data/waste.db) generates automatically. To rebuild manually: Windows: ...
Run Automated Tests	To run all unit tests: Windows: .venv\Scripts\python.exe -m pytest macOS/Linux: .venv/bin/pyt...
Windows Path Error	If you see 'Filename too long', move the folder to a short path such as C:\Projects\.

Tip: Refer to the included README.md and docs/methodology.md in the project folder for comprehensive documentation.